

MASTER'S COMPREHENSIVE EXAMINATION

Part I

Theory (90 minutes)

March 5, 2005

Instructions: Answer all four questions. Open book and open notes.

1. Let the joint mass function of the random vector (U, V) be as follows:

$\begin{matrix} u \\ v \end{matrix}$	0	1
0	.17	.03
1	.35	.02
2	.30	.13

- (a) Find the covariance of (U, V) .
- (b) Evaluate the c.d.f. $F(u, v)$ at $(u = 1, v = 1.5)$.
- (c) Find the conditional probability that $V = 2$, given $U = 1$.

2. Let X_1, X_2, X_3 be independent Poisson random variables with parameters 1, 2, 3, respectively.

- (a) Find the variance of $X_1 + 2X_2 - 8X_3 - 2$.
- (b) Find the second moment of X_3 .

3. We seek the difference in probabilities of hitting a home run for Mark McGuire and Sammy Sosa. The difference is $p_m - p_s$. Based on data for 1998, Mark hit 70 homers in 600 at bats while Sammy hit 65 (I think) in 690 bats. Find a 99% confidence interval for the difference.

4. Let X_1, X_2 be independent $N(\mu, 4)$. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$ by rejecting when $\bar{X} > C_\alpha$. Find C_α and find the power of the test when $\mu = 1$. Set $\alpha = .05$.